

CU Online — Chandigarh University

MCA Final Year Project — Milestone Report

MILESTONE 5 REPORT

Identify Cheapest Deployment Options

Analysing cost-per-workload data to determine optimal provider allocation by workload type

Project Title: Multi-Cloud Cost Comparison and Optimization Analytics Dashboard

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Milestone 5: Identify Cheapest Deployment Options

This Milestone involved a systematic analysis of the five-year cost and efficiency dataset — now accessible through the interactive dashboard — to identify the cheapest cloud provider for each of the three primary workload categories modelled in the project: analytics and batch processing workloads, production API and managed service workloads, and enterprise identity and integration workloads. This milestone translates the descriptive cost trend analysis into a prescriptive, workload-categorised deployment recommendation.

5.1 Workload Category Analysis

5.1.1 Analytics and Batch Processing Workloads

Analytics workloads — including data pipeline execution, business intelligence query processing, machine learning training, and batch data transformation — are characterised by high CPU and memory utilisation that runs continuously (24/7) rather than sporadically. This workload profile directly benefits from GCP's sustained-use auto-discount mechanism, which applies automatically when a resource runs for more than 25 percent of a billing month. At 100 percent monthly utilisation (which is typical for always-on analytics pipelines), GCP's sustained-use discount curve delivers a net 30 percent reduction from on-demand pricing — without any advance commitment or reservation required.

Additionally, GCP BigQuery — the serverless analytics engine — offers 10 GB of free storage and 1 TB of free query processing per month, making it particularly cost-effective for analytics workloads that can be structured as BigQuery queries rather than as always-on compute instances. The five-year trend analysis in the dashboard confirms GCP's analytics cost advantage: its cost-per-user declined 46 percent over the study period (from \$35.86K to \$18.67K), the steepest improvement of any provider, indicating that GCP's analytics infrastructure delivers better value per user at scale.

Conclusion for analytics workloads: GCP is the recommended provider. The five-year cost advantage (\$207.5M vs AWS's \$308.3M for equivalent workloads) combined with the steepest efficiency improvement rate makes GCP the optimal choice for all analytics, batch processing, and data pipeline workloads.

5.1.2 Production API and Managed Service Workloads

Production API workloads — services that must respond to user requests with low latency, scale rapidly under demand spikes, and integrate with a broad ecosystem of managed services (caches, queues, databases, CDN) — are best served by AWS. AWS's 200+ service catalogue includes managed services that have no direct equivalent on GCP or Azure: Lambda@Edge (for CDN-integrated serverless execution), Amazon Aurora (the highest-performance managed relational database), ElastiCache (managed Redis/Memcached), and CloudFront (the largest commercial CDN network). Architecting a production API system without these services on AWS would require significant additional engineering effort that would offset any cost savings from using a cheaper provider.

The five-year data confirms that AWS maintains consistent cost premium — \$308.3M versus GCP's \$207.5M — but this premium is partially justified by the breadth of services included. For workloads that specifically require AWS-unique services, the cost comparison must be assessed against the engineering cost of building equivalent capabilities on a cheaper platform.

Conclusion for managed API workloads: AWS is the necessary provider for workloads requiring AWS-specific managed services. Reserved Instance coverage of 60 to 70 percent of baseline compute can reduce effective AWS costs by 40 to 50 percent, narrowing the gap with GCP for the compute component of these workloads.

5.1.3 Enterprise Identity and Integration Workloads

Enterprise identity management — Active Directory, SSO (Single Sign-On), conditional access policies, MFA enforcement — is an area where Azure holds a specific and compelling advantage. Azure Active Directory (now Microsoft Entra ID) is the world's most widely deployed enterprise identity platform, used by more than 500,000 organisations. Its native integration with Microsoft 365, Windows Server, and the broader Microsoft enterprise software stack means that organisations already running Microsoft products can extend their existing identity infrastructure into Azure without the integration overhead that connecting an alternative identity platform would require.

For organisations with active Windows Server and SQL Server Software Assurance licensing, the Azure Hybrid Benefit can reduce the effective price of Azure VMs running these workloads by 40 to 60 percent, making Azure the lowest-cost option for these specific workload types despite its higher list prices relative to GCP.

Conclusion for enterprise identity workloads: Azure is the recommended provider. The Hybrid Benefit licensing advantage and Active Directory ecosystem integration make Azure unambiguously the optimal choice for identity, M365 integration, and Microsoft-stack workloads.

5.2 Workload-to-Provider Mapping Summary

Workload Category	Recommended Provider	Key Reason	Estimated Cost Position
Analytics, batch processing, data pipelines	GCP	Sustained-use auto-discounts; 46% cost/user improvement; BigQuery free tier	~27–30% cheaper than AWS
ML training, data transformation	GCP	Vertex AI; TPU access; GCS 15 GB always free	~27–30% cheaper than AWS
Production APIs, managed services	AWS	200+ service catalogue; Lambda@Edge; Aurora; CloudFront	Highest list cost; Reserved Instances reduce by 40–50%
CDN, global edge distribution	AWS	CloudFront — largest CDN; 400+ PoPs globally	Necessary for global reach
Enterprise identity, Active Directory	Azure	Hybrid Benefit 40–60% discount; native AD integration	~15% saving vs AWS for licensed workloads
Microsoft 365 integration	Azure	Native M365 connector; Entra ID ecosystem	Best-in-class integration
Windows Server workloads	Azure	Hybrid Benefit eliminates OS license cost	40–60% cheaper than on-demand for licensed VMs

Table M5.1: Workload-to-Provider Optimal Mapping